

ML PROJECT

About Automated Chilli Pepper Quality Grading

1.Opportunity Evaluation

Automated Mirchi Grading: Cutting Post- Harvest Loss with AI

- **Introduce the Problem**:Did you know the Mirchi industry loses millions due to ***inconsistent quality and slow grading***?Our Problem is Clear: Manual sorting is slow, subjective, and spreads spoilage.
- **Criteria**:Grading is visual, repetitive, inconsistent, making it a perfect opportunity for ML due to high feasibility and immediate economic value.



2. CRISP-DM Business Understanding

The Cost of Manual Grading

1. Define the Problem: The goal is to replace slow human inspection with a *real-time system*.

2. Define Success:

-> **Expected Business Impact:** A 15% reduction in Post-harvest losses attributed to misreading which lead to 10% increase in customer dissatisfaction within 6 months of deployment.

-> **Outcome Metric(Business Goal):** Reduction in Costs of Goods Sold due to minimised Material waste.

-> **Output Metrics(ML Performance):**

(i) Accuracy

(ii) Latency

(iii) Precision and Recall

3. Identify Factors:

(i) Chilli Variety

(ii) Camera Conditions

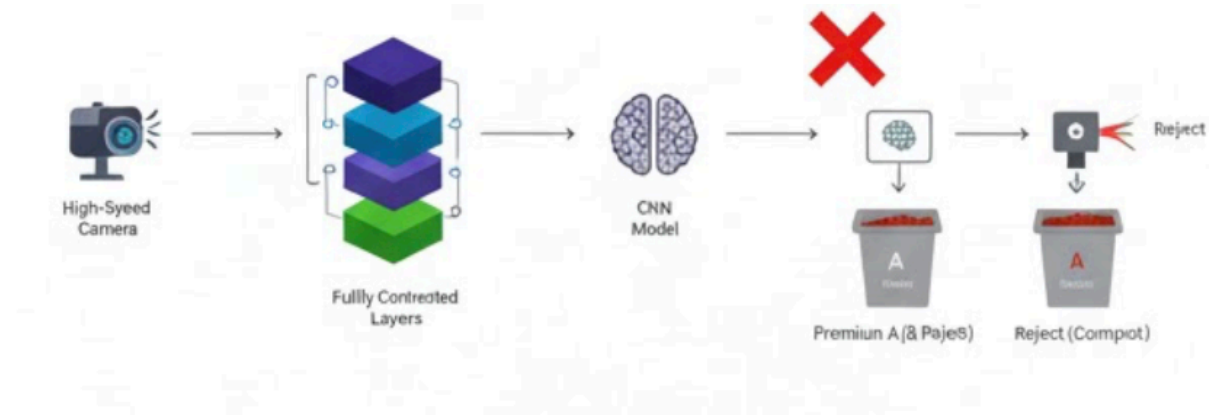
(iii) Seasonality

(iv) Moisture Content

3.Solution Validation Plan

Solution Concept and ML Solution

- **Describe the Solution concept how it uses ML:**Our Concept is the Automated Grading System. It uses Computer Vision in which it contains thousands of varieties of chillies images.
- **Describe iterative experiments to validate the concept:**Validation is iterative.We start with offline Testing and move to Pilot Deployment to Validate real-world performance before full launch.



4.ML System Design

Technical Design:Real-Time and Robust

1.Batch vs Real-Time Processing:

->**Decision:**Real-Time

->**Explanation:** We choose Real-time Processing because sorting must happen in milliseconds.

2.Human in Loop strategy:

->**Decision:**No Human-in-the-Loop for Primary Grading, but Human Oversight for Edge Cases.

->**Explanation:** We choose a Confidence Threshold.The model handles 95% of cases, low confidence images are being checked by humans which saves lot of time and collecting new data.

3.Model Retraining/Monitoring Strategy:

->**Decision:**Periodic Retraining(Monthly) with Automated Monitoring

->**Explanation:**We employ Automated Monitoring with Periodic Retraining to prevent Concept Drift caused by Seasonal Changes in defects.

5. Potential Risks in Productions

- 1. Training-Serving Skew:** Mitigated by Training of images from the actual factory camera/lighting setup.
- 2. Latency Failure:** Mitigated by choosing a light-weight, efficient CNN architecture to ensure inference time is always below the conveyor belt's required throughput.
- 3. Concept/Data Drift:** Mitigated by the continuous monitoring system ,which auto alerts us to retrain the model when a type of defect appears.

Conclusion

Our ML system shifts the Mirchi Business from subjective loss to objective profit. We offer a scalable, real-time solution that delivers better quality, greater efficiency, and a clear return on investment.